Residential Electricity Price Dynamics, Market Structure, and Underlying Drivers Across the PJM Interconnection, 2005–2025: An Empirical Analysis

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Abstract:

This paper assesses residential electricity prices in Ohio, Pennsylvania, Virginia, and New Jersey between January 2005 and December 2025 and asks how much of the movement in the available data can be attributed to any single factor. Data from the four states, all within the PJM Interconnection, are examined as a monthly panel of 1,008 observations. Although causal attribution was not achieved, the results illustrate how several critical supply and demand factors interrelate. Real residential prices rose in all four states, by 21.7 to 29.0 percent between the first and last twelve months of the sample. The PJM capacity price is the only variable with a significant positive coefficient in the pooled model, and the result holds across functional forms, although at the state level it is significant only in Pennsylvania and Virginia. Capacity prices reached the highest real levels in the sample in Ohio and Virginia in the 2025/26 delivery year, and the three most recent auctions since cleared at the regulated ceiling. Because PJM procures capacity against forecast load, data centers are a plausible driver, but other demand drivers cannot be excluded. Neither proxy examined in this study isolates data centers: commercial electricity sales (megawatt hours) has no significant effect in the main pooled specification, and information-sector GDP (millions of chained 2017 USD) has a significant negative effect. Despite mounting public concern about data centers and electricity demand, there is not sufficient evidence at present in the data to find a causal link. However, policy makers should consider requiring large facilities to report electricity use publicly to provide the data necessary to answer this question more clearly.

1. Introduction

In May 2025, residential customers in Ohio paid an average of 17.09 cents per kilowatt-hour for electricity. One year later that figure reached 19.52 cents, an increase of 14.2 percent (U.S. Energy Information Administration, 2026b). Households across the PJM Interconnection, the regional transmission organization (RTO) that operates the wholesale electricity market for all or part of 13 states in the eastern United States and the District of Columbia, have seen comparable increases.

Nationally, 78 percent of Americans say they are worried data centers are pushing their electricity prices higher, and in Virginia, home to the country's largest concentration of data centers, nearly three-quarters of voters blame the facilities directly for rising costs (Consumer Reports, 2026). That belief has moved into policy. In December 2025, Senator Elizabeth Warren and a group of congressional colleagues opened a formal investigation into whether large technology companies are shifting the cost of their data centers onto ratepayers (Warren, 2025).

By early 2026, more than 300 data-center-related bills had been introduced across over 30 states in a matter of weeks (MultiState, 2026). The evidence, however, is not clear cut. An industry-commissioned review of more than ten quantitative studies found no historical evidence that data centers have driven residential electricity cost increases under current rate structures, attributing recent increases instead to inflation, market design changes, and other supply-side factors (Energy + Environmental Economics, 2026).

This paper assesses residential electricity prices in Ohio, Pennsylvania, Virginia, and New Jersey between January 2005 and December 2025, and asks how much of that movement the available data can attribute to any single factor. The four states are examined as a monthly panel

of 1,008 observations. All four lie within the same grid, meaning the rules of the wholesale market are consistent across them while retail regulation, specific load growth, and zonal prices vary. Although our approach does not causally identify all the factors causing electricity prices to rise in recent years, descriptive analysis illustrates key correlations between several input and output measures.

2. Institutional Background

2.1 PJM's structure and footprint

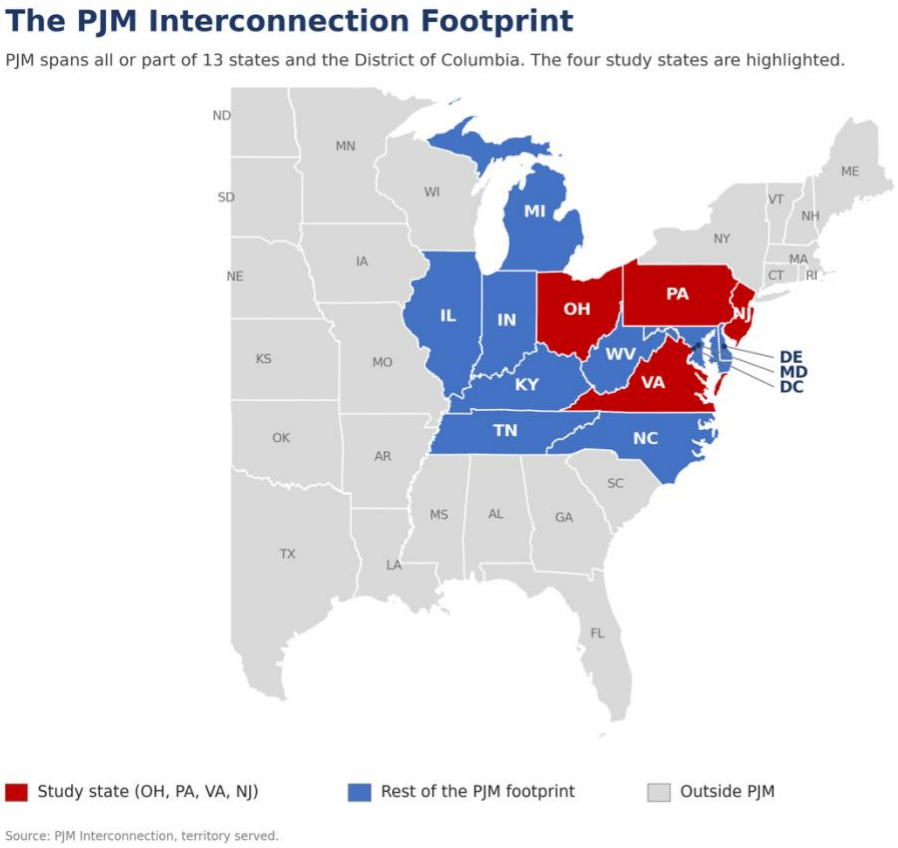
Understanding why residential electricity prices in these four states have risen requires understanding the market that determines the wholesale cost for electricity. The Pennsylvania-New Jersey-Maryland (PJM) Interconnection, commonly referred to by its acronym, is the largest regional transmission organization (RTO) in the United States. An RTO is an operator that runs the transmission grid and coordinates wholesale electricity markets on behalf of the utilities in its region, rather than owning the power plants or wires itself. PJM's footprint spans all or part of 13 states and the District of Columbia (Figure 1), covering most of the Mid-Atlantic and parts of the Midwest, including Pennsylvania, New Jersey, Maryland, Delaware, Ohio, Michigan, Illinois, Indiana, Virginia, West Virginia, North Carolina, Kentucky, and part of Tennessee. PJM became the nation's first fully functioning RTO in 2002, and PJM is regulated by the Federal Energy Regulatory Commission (FERC), which approves its market and operational rules (FERC, 2025).

This regional setting is why the analysis is structured as a four-state panel rather than a single-state case study. Because Ohio, Pennsylvania, Virginia, and New Jersey all sit within PJM,

they share the same wholesale market rules: the same capacity auction and the same process for setting local wholesale prices. Local wholesale prices and capacity prices are determined by market processes in different zones within PJM, so neither is uniform across the four states. Retail regulation and load growth differ by state as well. As a result, there are significant differences in household electricity prices across states.

Figure 1

Full PJM footprint, with the four states analyzed highlighted



The four states were selected because of their economic differences (e.g., which sectors are most important) and differences in retail electricity markets. Ohio and Pennsylvania are large, industrially diverse states with distinct regulatory histories. Virginia is a traditionally regulated

state sitting at the center of the nation's largest concentration of data-center demand. New Jersey, by contrast, is the smallest of the four geographically and carries one of the highest residential electricity prices in the PJM footprint.

2.2 The capacity market: history and design

Electricity markets pay for two different things. One is energy: the electricity generated and delivered in a given hour. The other is *capacity*: a payment to a generator simply for being available to run when called upon, whether or not it ends up running that day. PJM buys capacity separately from energy because keeping the grid reliable means having enough supply ready for the hours of peak demand, and energy prices alone may not earn generators enough to build it (Joskow, 2008). The supply of capacity must be built and maintained years in advance. The price PJM pays for that availability is the capacity price used throughout this analysis.

Before 2007, PJM bought capacity through a short-term arrangement that PJM itself later admitted produced low prices and high risk for investors, and that failed to attract enough new generation to meet future demand (PJM Interconnection, 2025a). PJM replaced it with a new system, approved by federal regulators in 2006 and first used in 2007, in which capacity for a given future year is bought three years ahead of time through a competitive auction (PJM Interconnection, L.L.C., 2006, 2007; Pfeifenberger et al., 2008). This history explains why the capacity price variable used throughout this analysis has no recorded values before June 2007 for any of the four states. The market itself did not yet exist.

In capacity auctions, PJM buys capacity from the cheapest available generators first and keeps buying, in order of ascending price, until it has secured enough to meet its reliability target. Every generator that clears the auction is then paid the same final price, set by the last,

most expensive unit needed to meet that target. Because PJM sets separate reliability targets for different zones of its territory, and must confirm that the power it buys can actually reach each zone given transmission limits, this final clearing price can differ from one zone to another even within the same auction (PJM Interconnection, 2025a).

The auction held in July 2024, covering the delivery year beginning June 2025, produced a sharp jump. Most of PJM's footprint cleared at $269.92 for each megawatt of capacity committed per day, up from $28.92 in the prior year's auction. Dominion, the zone covering most of Virginia, cleared even higher, at $444.26 (PJM Interconnection, 2025c, 2026e). Capacity prices are set separately by zone, such that Virginia's capacity price in this analysis diverges from the other three states.

The increase prompted a response from the states. In December 2024, Pennsylvania's governor filed a complaint with federal regulators challenging PJM's auction design, joined by the governors of Delaware, Illinois, Maryland, and New Jersey and nine state utility commissions. The resulting settlement capped prices between a ceiling and a floor for the following two auctions, set at $329.17 and $177.24 per megawatt-day for the 2026/27 delivery year. Regulators later extended these price limits through the 2029/30 auction, at a ceiling of $325.00 and a floor of $175.00. Every auction since has cleared at the ceiling.

These auction results suggest that electricity demand is likely to keep growing. The 2026 PJM forecast projects summer peak demand rising 3.6 percent per year over the next decade, although this was revised down from the previous year's forecast after PJM began screening large new connection requests more strictly before counting them (PJM Interconnection, 2026a). That downward revision means PJM itself has acknowledged that some of the demand behind recent auction price increases had not actually been confirmed. In the 2026/27 auction,

Dominion's price fell back in line with the rest of the market. In the 2027/28 auction, PJM failed to buy all the capacity it had determined the region needed, the first such shortfall in the system's history. A shortfall means PJM procured less than the 20 percent reserve margin it targets, so the system would run with slimmer reserves and greater risk, not that load goes unserved. The 2028/29 auction fell short again, and PJM is revising its Reliability Backstop Auction rules in response (PJM Interconnection, 2026e).

PJM attributes most of the projected demand growth to data centers, alongside electrification and growth in broader economic activity. A key mechanism connecting data centers to residential bills begins with these capacity markets. Announced data-center projects raise PJM's demand forecast, that forecast raises how much capacity PJM must buy, and the price of that purchase is passed on to customers. The connection runs through expected demand, not demand that has actually been metered and consumed.

2.3 Regulatory structure across the four states

Capacity prices and other wholesale transactions within PJM fall under federal jurisdiction, since they count as interstate commerce. Sales from a local utility to individual households do not. Instead, retail electricity rates are set separately by each state's own utility commission, municipality, or cooperative board (FERC, 2025). This split is why the analysis was conducted across four separate states rather than treated as one uniform market. The capacity price itself is set for the entirety of PJM, and is shared across states within a given zone, but the institutions that turn that wholesale cost into an actual household bill are specific to each state.

That translation from wholesale cost to household bill is what the terms regulated and deregulated describe. In a regulated state, a single utility both generates or procures the power

and sells it directly to customers at a rate a state commission sets, with no competing suppliers. In a deregulated, or restructured, state, residential customers can choose their own electricity supplier from a competitive retail market, while a single utility continues to own and operate the wires that deliver that power to homes. The wave of state restructuring laws passed in the late 1990s opened generation to competition while leaving the transmission and distribution wires as regulated monopolies, on the reasoning that building duplicate wires to the same homes would be wasteful even as competing to generate the power is not.

Ohio, Pennsylvania, and New Jersey are the three restructured states in this panel. Virginia is the regulated case. Pennsylvania restructured first, under the Electricity Generation Customer Choice and Competition Act of 1996. Ohio followed with Amended Substitute Senate Bill 3 in 1999, and New Jersey with the Electric Discount and Energy Competition Act the same year. Virginia passed its own restructuring act in 1999 and began phasing in retail choice but never completed the transition. In 2007 the General Assembly ended the program and returned the state to traditional regulation (Old Dominion Committee for Fair Utility Rates v. State Corporation Commission, 2017). Retail choice survives there only in narrow categories, including customers with demand above five megawatts (Va. Code Section 56-577(A)(3), (A)(5)), so ordinary Virginia households have been served by the same single utility for the entire period this analysis covers.

This difference matters because it influences how quickly and how directly a change in wholesale capacity price reaches a household bill. In a restructured state, residential customers buy generation from a competitive supplier whose price reflects wholesale costs, including capacity, and that price resets on a relatively short cycle. In a regulated state, the single utility recovers those same costs through rates the state commission approves, a slower administrative

process that can spread sudden increases across a longer period and a broader customer base. The wholesale signal is the same in both cases. What differs is the speed and the path by which it arrives to customers. Because only one of the four states is regulated, any comparison between the two structures identifies Virginia against the other three rather than a general effect of restructuring. Virginia also differs from the other three in ways beyond its regulatory status, including its generation mix and its concentration of data-center load, so the comparison is reported as suggestive rather than conclusive.

2.4 Data centers and load growth

Virginia and Ohio sit at the center of the nation's data center buildout. Virginia's Loudoun County, known as "Data Center Alley," hosts the densest concentration of data centers in the world, with an estimated 70 percent of global internet traffic either created in or passing through the county, a figure from state economic-development material rather than an independent traffic study (Virginia Economic Development Partnership, 2019). PJM's own forecasts reflect this concentration directly. PJM expects the Dominion zone, which covers most of Virginia, to see the largest absolute increase in summer peak demand of any zone in its territory between 2026 and 2030, driven largely by data-center load growth (PJM Interconnection, 2026a).

Central Ohio has emerged as a second, more recently developed data-center cluster, concentrated around New Albany and Licking County east of Columbus. Several major technology companies have built or announced facilities there since the early 2020s. Ohio's response shows that whether a state is regulated or deregulated does not determine how well it can respond to data-center-driven load growth. Even though Ohio is a restructured state, its public utilities commission has used its remaining authority over distribution rate design to adopt

a data-center-specific tariff, discussed further in Section 7.2 (Public Utilities Commission of Ohio, 2025).

New Jersey's relationship to data-center-driven load growth is less direct. Rather than hosting a comparable concentration of data centers itself, New Jersey's consistently high residential electricity prices are frequently attributed to its position within a part of the PJM footprint where transmission capacity is limited, though this analysis does not directly test that explanation. PJM's own materials describe how this kind of transmission congestion, meaning insufficient capacity to move power from where it is generated to where it is consumed, forces greater reliance on higher-cost local generation and raises delivered electricity costs in the affected area (PJM Interconnection, 2026d). Thus, New Jersey is not a state whose price trajectory can be explained by data-center demand within its borders. However, its consistently high prices are a useful comparison case because they appear to be driven by a different structural factor.

3. Data

3.1 Panel construction

The analysis uses a balanced monthly panel of four states, Ohio, Pennsylvania, Virginia, and New Jersey, covering January 2005 through December 2025. Each state contributes 252 monthly observations, for 1,008 observations in total.

Because the states share the PJM wholesale market but differ in retail regulation and load growth, cross-state comparison is useful for examining whether the relationships observed in the data are consistent across different state environments.

January 2005 was chosen based on data availability, since several series are not consistently reported at the state level before then. Beginning the sample in 2005 also provides several years of observations before the 2007–2009 financial crisis and captures the period preceding the later expansion of U.S. shale gas production.

3.2 Sources

Residential and commercial electricity data come from the U.S. Energy Information Administration (2026b). Capacity auction clearing prices and wholesale hub prices come from PJM Interconnection's public market data (PJM Interconnection, 2026b). State GDP by industry comes from the Bureau of Economic Analysis (BEA) Regional Economic Accounts (U.S. Bureau of Economic Analysis, 2026b). Heating and cooling degree days come from the National Centers for Environmental Information (2026). Where a series is published quarterly, its value is held constant across the quarter's three months.

3.3 Price deflation

All monetary series are deflated to constant 2017 dollars using the BEA Personal Consumption Expenditures Price Index (PCEPI; U.S. Bureau of Economic Analysis, 2026a). Deflation is important when making comparisons in prices across time. For instance, in this study we compare historical capacity prices with current capacity prices. Using the PCEPI, we correct for the effects general inflation could have on these capacity prices. Nominal prices thus appear only where explicitly labeled.

3.4 Variables

The variables used in the analysis are shown in Table 1. The dependent variable throughout the analysis is the monthly average residential electricity price, measured in cents per

kilowatt-hour. Across the full panel, the average is 13.29 cents per kilowatt-hour, with values ranging from 8.95 to 19.94 cents. There are persistent differences across states: New Jersey has the highest average residential price at 15.98 cents per kilowatt-hour, while Virginia has the lowest at 11.36 cents.

Table 1
Descriptive Statistics for the Panel, January 2005 to December 2025

Variable	*N*	Mean	SD	Minimum	Maximum
Residential price (cents/kWh)	1,008	13.29	2.05	8.95	19.94
Capacity price ($/MW-day)	892	118.19	63.37	17.22	350.61
Predicted natural gas price ($/Mcf)	1,008	15.29	4.97	8.14	29.73
Regional hub LMP ($/MWh)	1,008	42.51	20.61	13.95	151.96
Commercial sales (million MWh)	1,008	3.79	0.78	2.27	8.16
Information-sector GDP (billions 2017 $)	1,008	26.07	11.11	10.92	50.14
Manufacturing GDP (billions 2017 $)	1,008	70.09	24.07	38.67	109.33
Heating degree days	1,008	419.95	389.75	0.00	1,403.00
Cooling degree days	1,008	78.82	115.45	0.00	454.00

Note. Panel of four PJM states (Ohio, Pennsylvania, Virginia, New Jersey), 252 monthly observations each. All monetary values are deflated to constant 2017 dollars using the BEA Personal Consumption Expenditures Price Index (PCEPI). The capacity price is available from June 2007, when PJM's capacity market began, so its 892 observations reflect the absence of the market before that date rather than missing data. Information-sector and manufacturing GDP are reported quarterly and held constant across each quarter's three months.

The capacity price is the clearing price from PJM's annual Base Residual Auction for the capacity zone assigned to each state, measured in dollars per megawatt-day. This series begins in June 2007, when the current capacity market structure began, leaving 892 state-month observations for models that include capacity price. The 116 state-month observations before June 2007 are unavailable because the capacity-price series does not exist for that period, rather

than because of missing data within the sample. Accordingly, all models that include capacity price are estimated only for June 2007 onward.

The remaining variables serve as controls: natural gas price, regional wholesale hub price, manufacturing GDP, and heating and cooling degree days. Because natural gas prices are endogenous, we create a separate model for natural gas prices and use the resulting predicted natural gas price in the analysis (Section 3.4.1). The two variables used as data-center proxies, commercial electricity sales and information-sector GDP, are discussed separately in Section 3.6.

3.4.1 First-stage natural gas price regression

The predicted natural gas price used throughout the analysis is the fitted value from a first-stage regression of each state's residential natural gas price, deflated to constant 2017 dollars, on three national, supply-side instruments: the Henry Hub spot price, the West Texas Intermediate (WTI) crude oil price, and the volume of U.S. liquefied natural gas exports, each likewise deflated (U.S. Energy Information Administration, 2026c, 2026d, 2026e). Each instrument is a national series set independently of any single state's retail market, which is the basis for treating them as exogenous. The regression also controls for heating and cooling degree days, calendar-month indicators, and a linear time trend. It is estimated separately for each state, consistent with the per-state specification used elsewhere, with standard errors made heteroskedasticity- and autocorrelation-consistent using a 12-month lag. A rig-count sensitivity check was tested and is reported in Appendix F; it was not adopted into the specification shown in Table 1b below.

Table 1b
First-Stage Regression of Residential Natural Gas Price, Estimated Separately by State

Variable	Ohio	Pennsylvania	Virginia	New Jersey
Henry Hub ($/MMBtu)	0.4183*	0.3814***	0.5881***	−0.1167
	(0.1729)	(0.0551)	(0.0449)	(0.1288)
WTI oil ($/barrel)	−0.0054	−0.0003	0.0004	0.0150
	(0.0142)	(0.0072)	(0.0056)	(0.0133)
LNG exports (Bcf)	0.0068	0.0103***	0.0086***	0.0189***
	(0.0035)	(0.0018)	(0.0015)	(0.0033)
Joint F(3,234)	12.649***	41.520***	112.275***	33.313***
R^2	.819	.924	.933	.782
N	252	252	252	252

Note. Dependent variable is each state's residential natural gas price in dollars per thousand cubic feet, deflated to constant 2017 dollars (N = 252 per state). Standard errors, in parentheses, are heteroskedasticity- and autocorrelation-consistent (HAC) with a 12-month lag. LNG exports are measured in billions of cubic feet (Bcf); the underlying series is in millions of cubic feet, so those coefficients are scaled by 1,000. Heating degree days, cooling degree days, calendar-month indicators, and a linear time trend are included but omitted from the table. The joint F-test evaluates the three excluded instruments together against the conventional weak-instrument rule-of-thumb value of 10; all four joint tests have $p < .001$. Ohio's LNG coefficient has $p = .054$. Fitted values from these four regressions are the predicted natural gas price used throughout the rest of the analysis.
* $p < .05$. ** $p < .01$. *** $p < .001$.

The three instruments are jointly strong predictors of residential natural gas price in every state: the joint F-statistic exceeds 12 in all four states and 100 in Virginia, well above the conventional weak-instrument threshold of 10. Individually, Henry Hub is a significant positive predictor in Ohio, Pennsylvania, and Virginia, as expected, but not in New Jersey, where its coefficient is negative and insignificant. LNG exports is positive in all four states and significant in three, marginally so in Ohio ($p = .054$), making it the most consistent instrument individually. WTI oil is not independently significant anywhere, which is expected given its correlation with the other two; it still contributes to the joint relevance of the set. The resulting fitted values are the predicted natural gas price used in Table 3, Table 3b, and Table 5.

3.5 Capacity zone assignment

PJM sets capacity prices by capacity zone rather than by state. For this analysis, each state is assigned the zone that covers most of its load (Table 2).

Table 2
PJM Capacity Pricing Zone Assigned to Each State

State	Capacity zone	Note
Ohio	RTO	Clears at the region-wide price
Pennsylvania	MAAC	Spans more than one zone; assigned its closest single match
Virginia	Dominion	Cleared at the region-wide price through 2024/25; separated in 2025/26
New Jersey	EMAAC	Eastern MAAC, transmission-constrained

Note. PJM sets capacity prices by zone rather than by state, so each state is assigned the zone covering the majority of its load. Because Virginia's Dominion zone cleared at the region-wide RTO price in every delivery year except 2025/26, Ohio and Virginia carry identical capacity prices in all but one year of the sample.

Two features of this assignment are critical for the analysis. Pennsylvania spans multiple capacity zones, so assigning it to MAAC is an approximation rather than a perfect representation of all Pennsylvania load. Virginia presents a different issue. The Dominion zone cleared at the region-wide RTO price in every delivery year through 2024/25, so Ohio and Virginia have identical capacity prices in all but one year of the sample. Thus any inferences about the implications of capacity prices are derived largely from intertemporal variation rather than variation across states.

3.6 Data-center proxies

No publicly available dataset reports electricity consumption attributable specifically to data centers at the state and monthly level. This is not simply a gap in the sources used for this study. To the extent that data-center electricity demand is measured at this level of geographic and temporal detail, it is not publicly reported as a standalone series. Any analysis of data-center demand at this resolution must therefore rely on proxies.

Two proxies are available consistently across all four states for the full sample. Commercial electricity sales capture where large new loads have physically appeared, since data centers are generally classified as commercial customers for electricity billing purposes. Information-sector GDP captures the broader industry classification that includes data processing and hosting. The GDP series is reported quarterly (Section 3.2).

Neither proxy is specific to data centers, and the two have different limitations. Commercial electricity sales identify the relevant customer category and geography but not the specific source of demand. The series aggregates all commercial electricity customers in a state, so growth associated with data centers cannot be separated from growth in offices, warehouses, retail, health care, and other commercial activities. Information-sector GDP is more closely aligned with the industry in which data processing and hosting are classified, but it is also broader than data centers. The sector includes publishing, telecommunications, broadcasting, and software, among other activities. It also grew at nearly identical rates in Ohio and Virginia over the sample (Section 5.3).

These limitations are central to the interpretation of the results below. The coefficients on both proxies cannot be interpreted as a data-center-specific relationship because neither variable isolates data-center activity. This limitation also motivates the recommendation for more detailed data on data-center electricity consumption.

3.7 Key assumptions

Several measurement and modeling decisions are important for interpreting the results. These choices are summarized here rather than repeated throughout the analysis.

- Monetary control variables are expressed in constant 2017 dollars to remove the effect of general price changes over time (Section 3.3).
- Each state is assigned a single capacity zone for the analysis: Ohio to the RTO zone, Pennsylvania to MAAC, New Jersey to EMAAC, and Virginia to Dominion (Section 3.5).
- Capacity prices are reported by delivery year, which runs from June through May. Because the sample ends in December 2025, only June through December 2025 of the 2025/26 delivery year is included.
- Information-sector and manufacturing GDP are reported quarterly and held constant within each quarter (Section 3.2).
- Neither data-center proxy directly measures data-center activity. Commercial electricity sales capture a broad category of commercial demand, while information-sector GDP covers a broader set of industries that includes data processing and hosting (Section 3.6).
- The panel contains only four states. This limits the reliability of cluster-robust inference in the pooled model and is considered when interpreting statistical significance (Section 4.4).

These choices do not prevent the analysis from identifying descriptive relationships in the data, but they place limits on how precisely individual coefficients can be interpreted. In particular, the capacity-zone assignments and data-center proxies are measurement decisions that should be kept in mind when interpreting the state-level and pooled results.

4. Methodology

4.1 Approach

This analysis is mostly descriptive, assessing which observable factors are associated with residential electricity prices across four PJM states. It does not attempt to identify causal effects. With the exception of natural gas prices, for which we do create an instrument, no other instruments, natural experiments, or exogeneity assumptions are used. As a result, the coefficients reported below should not be interpreted for causality. The goal is to establish which relationships are present in the data and, just as importantly, which questions the available data cannot answer.

Two estimations are used. The first pools all four states to identify which factors are associated with residential prices across the panel as a whole. The second estimates the same model separately for each state to examine whether the two available data-center proxies behave differently in states with different levels of data-center activity.

4.2 Pooled panel specification

The pooled model regresses the monthly residential electricity price on the capacity price and a set of controls from June 2007 through December 2025, the period in which PJM's capacity market has operated (N = 892). The controls are predicted natural gas price, regional hub price, manufacturing GDP, information-sector GDP, commercial electricity sales, heating and cooling degree days, and a linear time trend.

The model includes state fixed effects and calendar-month fixed effects. State fixed effects account for characteristics that are constant within each state over the sample, including its average residential electricity price, regulatory structure, generation mix, and capacity-zone

assignment. Calendar-month fixed effects account for seasonal patterns that are common across the four states. The coefficients are therefore estimated from within-state changes over time after controlling for these state-specific differences and common seasonal patterns.

The pooled correlation between capacity price and residential electricity price is 0.41 across all states, which is higher than the within-state correlation in Ohio (0.10), Pennsylvania (0.05), and Virginia (0.23), though not New Jersey (0.43). For most states, this suggests that part of the pooled association comes from differences in price levels across states rather than the two variables moving together within a state. The fixed effects remove this cross-state component, which is why the fixed-effects regression coefficient, rather than the raw correlation, is used as the main evidence in Section 5.1.

Variables measured in large units are rescaled so that the coefficients are easier to interpret. Manufacturing and information-sector GDP are measured in billions of chained 2017 dollars, while commercial electricity sales are measured in millions of megawatt-hours. Rescaling changes the units of the reported coefficients but does not change the underlying relationships or statistical significance of the results.

As a robustness check, the pooled model is also estimated in log-log form, with every continuous regressor expressed as a natural logarithm, and in semi-log form, with only the dependent variable logged. Both are reported alongside the levels specification in Section 5.1.1.

4.3 Per-state specification

To test whether the data-center proxies behave differently across states, the same specification is estimated separately for each state (N = 223 per state). Each model includes the two data-center proxies alongside capacity price, predicted natural gas price, heating and cooling

degree days, a linear time trend, and calendar-month fixed effects. Because each model contains only one state, state fixed effects are not applicable.

The same specification is used for all four states rather than adjusting the model to fit each state individually. Although state-specific specifications could improve the fit of individual models, they would make the resulting coefficients less directly comparable. Using a common specification keeps the variables and controls consistent across states, which is important because the purpose of these regressions is to compare how the data-center proxies behave across states.

4.4 Inference

Standard errors in the pooled model are clustered by state. Cluster-robust inference generally relies on a reasonably large number of clusters, while this panel contains only four states. With so few clusters, cluster-robust standard errors and the resulting significance tests can be unreliable. The statistical significance of the pooled results should therefore be interpreted cautiously.

Standard errors in the per-state models are heteroskedasticity- and autocorrelation-consistent (HAC), using a 12-month lag to account for the serial correlation expected in monthly price series. This allows the standard errors to remain robust to both changing error variance and correlation across observations over time.

Because both models include a linear time trend and calendar-month fixed effects, a variable must be associated with residential prices beyond a steady trend and the common seasonal cycle to produce a statistically distinguishable coefficient. Variables that simply

increase over the sample period do not necessarily produce a significant coefficient simply because they trend upward alongside residential prices.

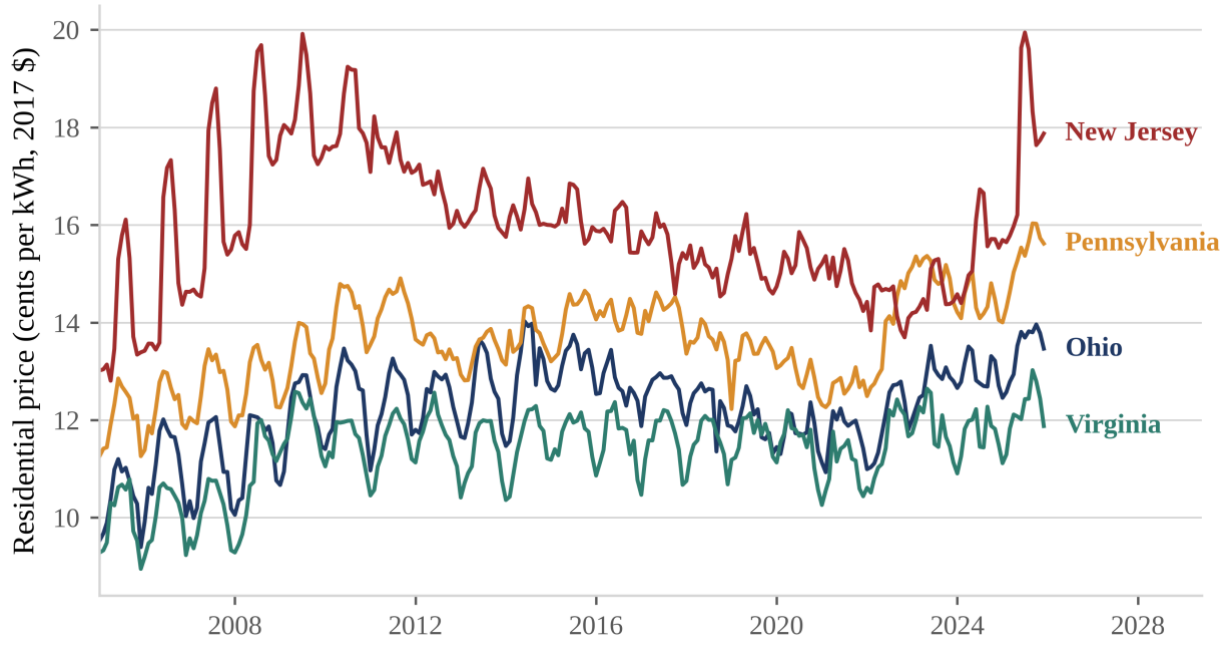
5. Results

5.1 What moved residential prices

Residential electricity prices rose in real terms in all four states (Figure 2). Comparing the first twelve months of the sample to the last, and holding prices in constant 2017 dollars, Ohio rose from 10.38 to 13.38 cents per kilowatt-hour (29.0 percent), Pennsylvania from 12.02 to 15.26 (26.9 percent), New Jersey from 14.03 to 17.50 (24.7 percent), and Virginia from 9.96 to 12.12 (21.7 percent). Because these figures are deflated, the increase is not a product of general inflation.

Figure 2

Residential electricity prices by state, January 2005 to December 2025, in constant 2017 dollars



To identify which factors are associated with changes in residential electricity prices, a pooled panel regression was estimated across all four states over the period in which PJM's capacity market has operated, from June 2007 through December 2025 (N = 892). The model includes state fixed effects to control for differences across states, such as regulatory structure and generation mix, as well as calendar-month fixed effects to control for seasonal patterns common across the four states. Standard errors are clustered at the state level. The model estimates relationships using within-state variation after accounting for differences across states and common seasonal effects.

Table 3

Pooled Panel Regression of Residential Electricity Price on Capacity Price and Controls

Variable	Coefficient	SE	*p*
Capacity price ($/MW-day)	0.0032**	0.0010	.002
Predicted natural gas price ($/Mcf)	−0.0070	0.0290	.810
Regional hub LMP ($/MWh)	0.0005	0.0044	.904
Manufacturing GDP (billions 2017 $)	0.0115	0.0195	.555
Information-sector GDP (billions 2017 $)	−0.0347***	0.0066	< .001
Commercial sales (million MWh)	0.3004	0.2207	.173
Heating degree days	−0.0006	0.0004	.102
Cooling degree days	−0.0019*	0.0009	.043
Time trend (months)	0.0023	0.0041	.566
State fixed effects	Yes		
Month fixed effects	Yes		
N	892		
R^2	.825		

Note. Dependent variable is the monthly residential electricity price in cents per kilowatt-hour, deflated to constant 2017 dollars. The sample covers June 2007 through December 2025, the period in which PJM's capacity market has operated. Standard errors are clustered by state; with four clusters, cluster-robust standard errors may be understated. State and calendar-month fixed effect coefficients are estimated but omitted from the table.

* p < .05. ** p < .01. *** p < .001.

The capacity price has a positive coefficient, as expected, illustrating a positive correlation between capacity prices and residential electricity prices. Its coefficient is statistically significant (+0.0032, $p = .002$). Holding state characteristics and seasonality constant, a one-dollar-per-megawatt-day increase in the capacity price is associated with an increase of roughly 0.0032 cents per kilowatt-hour in the residential electricity price.

Surprisingly, predicted natural gas prices are not significantly different from 0, and the coefficient is negative. This variable is potentially picking up the long-term trend in natural gas prices, which has been downward, versus the upward long-term trend in electricity prices. In the linear model, many of the other variables are also insignificantly different from 0, e.g., the regional hub price ($p = .904$), manufacturing GDP ($p = .555$), commercial electricity sales ($p = .173$), and the linear time trend ($p = .566$). Cooling degree days are statistically significant ($p = .043$), while heating degree days are not ($p = .102$). However, both have relatively small coefficients and are included primarily as controls to explain the annual cyclical price variation rather than as explanations for the longer-term increase in residential prices. These results imply that several commonly cited factors considered in this model, including fuel costs, wholesale energy prices, and industrial activity, do not have statistically significant relationships with residential prices once state and seasonal differences are controlled.

One coefficient does not have a straightforward interpretation. Information-sector GDP is statistically significant at $p < .001$ but has a negative coefficient, implying that increases in information-sector GDP are associated with lower residential electricity prices. This relationship is difficult to reconcile with the expected relationship between growth in the information sector and electricity demand. However, the timing of financial flows through the information sector, data center development and implementation, and energy price effects is complicated. It is

entirely likely that economic activity related to data centers has not yet emerged significantly in state-level income accounts, while strong expectations about future growth, as evidenced by the rising capacity prices, have already begun influencing electricity prices. Section 5.3 examines this issue further, since information-sector GDP is one of the proxies used to measure data-center activity.

The model explains a substantial share of the variation in residential electricity prices (R^2 = 0.825), although much of this explanatory power is attributable to the fixed effects rather than the variables of primary interest. Because the clustered standard errors rest on only four clusters, the statistical significance of the capacity price coefficient should be interpreted with caution (Section 4.4).

5.1.1 Robustness to functional form

Table 3 is reestimated in two further functional forms to test whether the results depend on the linear specification. The log-log model expresses the dependent variable and every continuous regressor as a natural logarithm, so its coefficients are elasticities: the percent change in price per one percent change in the regressor. The semi-log model logs only the dependent variable, so its coefficients are semi-elasticities; multiplied by 100, they give the percent change in price per one-unit change in the regressor. Heating and cooling degree days are exactly zero in a large share of the sample — 79 of 892 observations for heating degree days, 391 for cooling degree days — and the log of zero is undefined, so they enter the log-log model as $\ln(x + 0.001)$ and the semi-log model in levels. The time trend and the state and calendar-month fixed effects are unchanged throughout. Both models use the same sample as Table 3 (N = 892).

Table 3b
Pooled Panel Regression, Log-Log and Semi-Log Specifications

	Log-log (elasticity)			Semi-log (semi-elasticity)		
Variable	Coef.	SE	*p*	Coef.	SE	*p*
Capacity price	0.01432*	0.00676	.034	0.00023***	0.00006	< .001
Predicted natural gas price	0.03364	0.05669	.553	0.00016	0.00238	.946
Regional hub LMP	−0.00669	0.01265	.597	−0.00013	0.00035	.710
Manufacturing GDP	0.03272	0.07821	.676	0.00057	0.00128	.655
Information-sector GDP	−0.13377***	0.02398	< .001	−0.00261***	0.00048	< .001
Commercial sales	0.15812*	0.06648	.018	0.02219	0.01451	.127
Heating degree days	0.00113	0.00196	.565	−0.00003	0.00004	.406
Cooling degree days	0.00054	0.00043	.217	−0.00012	0.00007	.116
Time trend	0.00043*	0.00020	.029	0.00022	0.00025	.386
State fixed effects	Yes			Yes		
Month fixed effects	Yes			Yes		
N	892			892		
R^2	.838			.837		

Note. Dependent variable is the residential electricity price in cents per kilowatt-hour, logged in both specifications. Log-log coefficients are elasticities; semi-log coefficients are semi-elasticities (multiply by 100 for the percent change in residential price). Heating and cooling degree days enter the log-log model as ln(x + 0.001), since each is zero in many months, and the semi-log model in levels. The time trend is in levels in both models. Standard errors are clustered by state; with four clusters, cluster-robust standard errors may be understated. State and calendar-month fixed effects are included but omitted from the table. Units: capacity price, $/MW-day; predicted natural gas price, $/Mcf; regional hub LMP, $/MWh; manufacturing and information-sector GDP, billions of 2017 dollars; commercial sales, millions of MWh; time trend, months.
* p < .05. ** p < .01. *** p < .001.

Capacity price and information-sector GDP carry the same sign and remain significant in both additional specifications, so neither result depends on the linear functional form. Commercial electricity sales is insignificant in levels (p = .173) but significant in log-log form (p = .018), implying its relationship with price is proportional rather than additive — a difference in functional form, not a contradiction between specifications. Predicted natural gas price, the regional hub price, and manufacturing GDP remain statistically indistinguishable from zero throughout. Appendix B reports a broader version of the pooled model that adds industrial sales, and Appendix C reports a random forest check; neither changes these conclusions.

5.2 Why capacity prices matter

Section 5.1 identified capacity price as the variable most clearly associated with residential electricity prices. To understand the pathway for these prices, which reflect expectations about future electricity demand in the region, consider how PJM pays generators.

Generators are paid in two ways. The first is a payment for electricity actually produced. The second is payment for committing to be available during the hours of highest system demand. This capacity is procured through an annual auction, with the cost recovered from load-serving entities and ultimately passed on to retail customers. Every household in a capacity zone therefore pays a share of the capacity cost regardless of its own electricity consumption.

The important feature here is how PJM determines how much capacity to procure. PJM does not buy capacity based on metered consumption. Instead, it buys against a forecast of future peak demand, and that forecast incorporates large loads that have been announced or have entered the interconnection queue. A facility that has not yet been built, and may never be, can therefore increase the amount of capacity PJM needs to procure and potentially raise the price paid by customers in that zone. The connection between announced large loads (e.g., future data centers) and household electricity bills is therefore anticipatory rather than based on actual consumption.

The link between future expected data center demand, capacity prices, and current electricity prices has implications for what the data in this paper can and cannot show. A variable measuring realized demand will not necessarily capture a cost driven by expected future demand, and the two measures do not have to move together. This helps explain why a demand-side measure may show little or no relationship with residential prices while the capacity price does. It also means that the weak results for the data-center proxies in Section 5.3 should not be read as

evidence that large loads do not affect residential prices: the proxies measure realized demand, and the effect may run through expected demand.

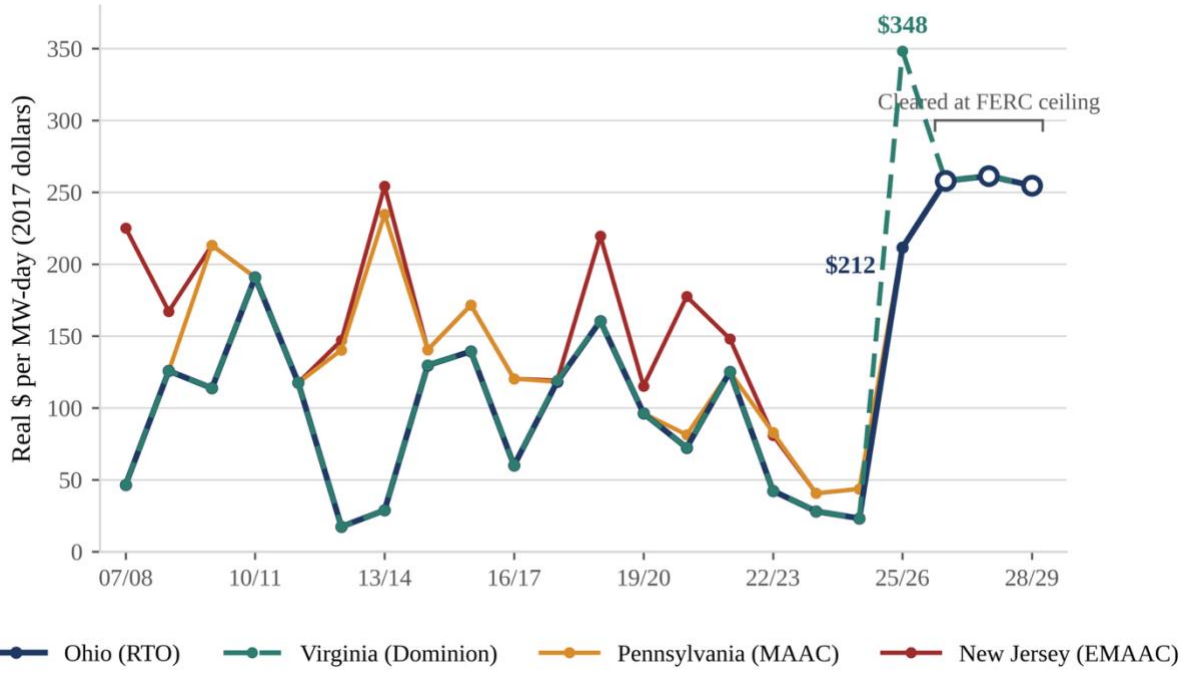
PJM's forecasts have also changed as the assumptions underlying them have been revised. PJM revised its 2026 forecast downward after applying stricter screening to large interconnection requests, indicating that some of the demand included in earlier forecasts had not been confirmed. The subsections below examine how capacity prices have changed over time and what the most recent auctions reveal about the relationship between capacity demand and prices.

5.2.1 Capacity prices have reached historically high levels

Capacity prices are deflated to constant 2017 dollars and are reported by delivery year (June through May), since a calendar-year average would blend two auctions (Figure 3). The 2025/26 RTO price of $212 in real terms (2017) corresponds to a nominal clearing price of $269.92.

Figure 3

PJM capacity auction clearing prices by delivery year, in constant 2017 dollars



In Ohio the 2025/26 price of $212 per megawatt-day is the highest real value in the sample, about 11 percent above the previous high of $191 in 2010/11. Virginia's $348 is 82 percent above its own prior high. Pennsylvania and New Jersey remain below their earlier peaks, so the current level is historically high in two of the four states rather than all four.

The series does not rise steadily, and its peaks differ by zone. The RTO zone peaks at $191 in 2010/11, while Pennsylvania and New Jersey peak in 2013/14 at $235 and $254 against $29 in the RTO zone that same year. Pennsylvania cleared above the region-wide price in ten of the eighteen delivery years before 2025/26 and New Jersey in fifteen, so zonal separation is routine; what distinguishes 2025/26 is that Virginia separated, having cleared at exactly the region-wide price in every prior year. Troughs fall as low as $17 in 2012/13, and the recent increase follows the lowest stretch in the sample, with Ohio and Virginia at $23 in 2024/25 immediately before rising.

Note that the 2025/26 delivery year is represented by June through December 2025, the months available before the sample ends, rather than a full twelve, and the Ohio–Virginia divergence rests on a single state-year, one that closes again in 2026/27 when all four zones clear at a common price.

5.2.2 Recent auctions have cleared at their regulatory ceiling

Three auctions have cleared for delivery years beyond the sample, and all three cleared at the applicable FERC-approved ceiling: $329.17 per megawatt-day for 2026/27, $333.44 for 2027/28, and $325.00 for 2028/29, nominal, with no zonal separation in any of them (Table 4). The next auction, for 2029/30, is scheduled for December 2026 and had not been held at the time of writing.

Table 4
PJM Base Residual Auction Outcomes, 2025/26 Through 2028/29

Delivery year	Clearing price (nominal $/MW-day)	Cleared at ceiling	Procurement vs. reliability requirement	Installed reserve margin
2025/26	$269.92	No	—	—
2026/27	$329.17	Yes	+139 MW	—
2027/28	$333.44	Yes	−6,517 MW	14.9%
2028/29	$325.00	Yes	−6,831 MW	14.7%

Note. Clearing prices are nominal, as published. The 2025/26 delivery year is the last in the estimation sample; the three subsequent auctions each cleared at the FERC-approved price ceiling established in Docket ER25-1357 and extended through 2029/30. Procurement is measured against the RTO reliability requirement and includes Fixed Resource Requirement commitments. PJM's target installed reserve margin is 20%. The 2029/30 auction was scheduled for December 2026 and had not been held at the time of writing. Values from PJM Base Residual Auction reports and accompanying press releases (PJM Interconnection, 2025a, 2025b, 2025c, 2026c, 2026e).

A price that clears at its cap provides less information than one that clears freely because the price ceiling limits the outcome. Several features suggest that the constraint was binding rather than incidental. Monitoring Analytics' simulation of the 2027/28 auction without the price ceiling produced a materially higher price (Monitoring Analytics, 2026). The 2027/28 auction was also the first in the market's history in which the entire RTO, including Fixed Resource Requirement areas, fell short of the reliability requirement. The shortfall was 6,517 megawatts, with the auction clearing at a 14.9 percent installed reserve margin compared with a 20 percent target. The 2028/29 auction fell short again, this time by 6,831 megawatts. PJM described this outcome as not unexpected given the shortfall in the previous auction. Two consecutive shortfalls under a binding price cap suggest that the market was experiencing scarcity that was not fully reflected in the cleared price.

PJM directly attributes much of the demand growth behind these auctions to data centers. Its 2026/27 auction materials report that forecasted peak load increased by more than 5,400 megawatts from the previous year, driven largely by data center expansion, electrification, and

economic growth (PJM Interconnection, 2025a, 2025c). The 2027/28 and 2028/29 auction reports also describe the continued addition of large loads to the load forecast used to determine the reliability requirement (PJM Interconnection, 2025b, 2026e). This provides evidence from the market operator that large-load growth, which PJM attributes largely to data centers, is becoming an important part of the demand being incorporated into PJM's forecasts. It is also consistent with the mechanism described above. However, these statements do not establish how much of the increase in forecasted demand is attributable specifically to data centers. Section 5.3 explains why that quantity cannot be directly recovered from the available public data.

5.3 The data-center proxies do not isolate activity

Two variables are available consistently across all four states as proxies for data-center activity: commercial electricity sales, which capture where large new load has physically arrived since data centers are billed as commercial customers, and information-sector GDP, which covers the industry classification containing data processing and hosting. Both were selected because they are the closest available measures, not because either is specific to data centers.

Table 5

Data-Center Proxies and Residential Electricity Price, Estimated Separately by State

	Information-sector GDP (per $B)			Commercial sales (per million MWh)				
State	Coef.	SE	*p*	Coef.	SE	*p*	*N*	R^2
Ohio	0.0738	0.0561	.188	1.3091***	0.2765	< .001	223	.705
Pennsylvania	−0.0511	0.0293	.081	1.0828*	0.5420	.046	223	.416
Virginia	0.0114	0.0218	.600	0.2820	0.1556	.070	223	.652
New Jersey	0.1477	0.0935	.114	−0.0340	0.4723	.943	223	.587

Note. Dependent variable is the monthly residential electricity price in cents per kilowatt-hour, deflated to constant 2017 dollars. One regression per state over June 2007 through December 2025, using a common specification that also includes the capacity price, the predicted natural gas price, heating and cooling degree days, a linear time trend, and calendar-month fixed effects. Standard errors are heteroskedasticity- and autocorrelation-consistent with a 12-month lag. Information-sector GDP is expressed in billions of chained 2017 dollars and commercial sales in millions of megawatt-hours, so coefficients give the change in price associated with a one-unit increase in each.

* p < .05. ** p < .01. *** p < .001.

The regressions were estimated separately for each state using the same specification (Table 5). Information-sector GDP is not statistically distinguishable from zero in any of the four states. Commercial electricity sales are statistically significant in Ohio and Pennsylvania, marginally significant in Virginia, and not significant in New Jersey.

The pattern of these results does not match the distribution of data-center load across the states. Pennsylvania, which does not have a comparable concentration of data centers, has a large and statistically significant commercial-sales coefficient. Virginia, which has one of the largest concentrations of data centers in the world, has the smallest positive coefficient of the three states where the coefficient is positive, and it does not reach conventional levels of statistical significance. If either variable were capturing data-center demand specifically, the results would be expected to show a stronger relationship in states with greater data-center activity.

The reason is largely determined by what each variable measures. Commercial electricity sales include every commercial customer in a state, including offices, warehouses, hospitals, and retail businesses. Therefore, a significant coefficient could reflect growth in ordinary commercial activity rather than data centers. Information-sector GDP has a similar problem. It includes data processing along with publishing, telecommunications, broadcasting, and software. It also grew at nearly identical rates in Ohio (133 percent) and Virginia (135 percent) over the sample, despite the large difference in data-center activity between the two states. A variable that grows at nearly the same rate in states with very different levels of data-center activity is not a reliable measure of data-center load.

These results show that the two proxies cannot answer the main question of this paper. The problem is primarily one of measurement rather than statistical power. A coefficient

estimated using either variable cannot be interpreted as the effect of data centers because neither variable isolates data-center activity. Increasing the sample size or changing the model specification would not solve this measurement problem. Section 7 returns to what additional data would be needed to directly measure the relationship between data-center development and residential electricity prices.

5.4 The capacity price by state

The per-state regressions in Table 5 also include the capacity price. Table 5b reports its coefficient. It is positive in all four states, as expected, but significant only in Pennsylvania (p = .044) and Virginia (p = .005). Pairwise correlations for each state are reported in Appendix E.

Table 5b
Capacity Price Coefficient, Estimated Separately by State

State	Coefficient	SE	*p*
Ohio	0.0006	0.0009	.515
Pennsylvania	0.0034*	0.0017	.044
Virginia	0.0020**	0.0007	.005
New Jersey	0.0045	0.0041	.269

Note. Coefficients are on the capacity price ($/MW-day) from the per-state regressions reported in Table 5, using the same specification (June 2007 through December 2025, N = 223 per state). Dependent variable is the monthly residential electricity price in cents per kilowatt-hour, deflated to constant 2017 dollars. Standard errors are heteroskedasticity- and autocorrelation-consistent with a 12-month lag, and p-values are from the normal distribution, as in Table 5.
* p < .05. ** p < .01. *** p < .001.

In Ohio the coefficient is close to zero (p = .515). In New Jersey it is the largest of the four but imprecisely estimated (p = .269), so it cannot be distinguished from zero or from the pooled estimate of 0.0032 (Table 3). The pooled result therefore does not hold uniformly across states. Virginia's coefficient does not depend on the months in which the Dominion price separated from the RTO price (Section 3.5): excluding June through December 2025 leaves it similar in size but less precisely estimated (0.0023, p = .054).

6. Discussion

6.1 What the results support, and what they do not

The capacity price is the only variable with a significant positive coefficient in the main specification (Table 3), and it remains positive and significant in the log-log and semi-log forms (Table 3b). Predicted natural gas prices, the regional wholesale price, and manufacturing GDP are statistically indistinguishable from zero throughout.

Although this analysis does not causally identify the capacity price's effect on residential prices, it illustrates a correlation that survives the functional forms tested. It does not hold uniformly across states: the per-state coefficient is significant only in Pennsylvania and Virginia (Table 5b). Natural gas is the only instrumented variable (Section 3.4.1), so every coefficient should be read as an association, not an effect.

Of the two data-center proxies, commercial electricity sales is insignificant in levels but significant in log form (Table 3b), and information-sector GDP is significant but with the wrong sign. Section 6.5 argues this reflects a timing problem in the proxy rather than evidence against the data-center channel.

6.2 Why capacity is a plausible channel

The capacity price is the one forward-looking price in the model, which is what makes it a plausible conduit for data centers. PJM procures capacity against a forecast of future peak load that includes large loads that are announced or in the interconnection queue, and the cost reaches every household in the zone through load-serving entities (Section 5.2). A data center can

therefore raise what households pay before it consumes a kilowatt-hour. PJM's auction reports are consistent with this: forecasted load behind the 2027/28 and 2028/29 reliability requirements rose by 5,249.9 and 1,374.5 megawatts, respectively, largely due to additional large loads (PJM Interconnection, 2025b, 2026e), and the three most recent auctions cleared at the price ceiling (Table 4).

The most direct evidence against a price-raising role for data centers is Watten et al. (2026), who instrument data-center siting and find that data centers may have lowered average retail rates from 2015 to 2024, because new demand spreads fixed grid costs over more kilowatt-hours. This complements the capacity channel rather than contradicting it. Their estimate captures realized load, while the capacity price responds to expected load. They note that the effect reverses if anticipated load materially exceeds realized consumption, since capacity built in advance is then paid for by fewer kilowatt-hours. PJM's downward revision of its 2026 forecast after stricter screening of large-load requests shows that gap is not hypothetical (Section 2.2). Their sample also ends in 2024, before the 2025/26 delivery year, when the capacity price rose from $28.92 to $269.92 per megawatt-day in nominal terms (PJM Interconnection, 2026e).

The two sets of evidence therefore describe different stages. Although this analysis cannot identify how much of the capacity price increase data centers explain (Section 6.3), it shows that capacity price is the variable that tracks residential prices, and it is the stage at which expectations rather than consumption set the bill.

6.3 Why the data-center proxies cannot answer the question

The proxies fail for a reason that more data or a better specification would not fix: neither one separates data centers from everything else in its category (Section 5.3). The clearest signal

is in Virginia, where commercial sales rose by 29.6 million megawatt-hours, or 55 percent, between 2019 and 2025, which the Energy Information Administration attributes mainly to data centers (U.S. Energy Information Administration, 2026a).

The state-level pattern shows the problem (Table 5). If commercial sales were picking up data centers, its coefficient should be largest where data centers are concentrated. It is instead largest in Ohio and Pennsylvania, where it is significant, and small and only marginally significant in Virginia ($p = .070$). Virginia illustrates the difficulty from the other side: its real residential price rose 3.6 percent between 2019 and 2025, the smallest increase of the four states, while its commercial sales grew the most. Because Virginia is the only regulated state, this comparison cannot separate load growth from rate structure (Section 6.4). The likelier reading of the Ohio and Pennsylvania coefficients is that commercial sales track general economic activity, which moves with prices for reasons unrelated to data centers.

This is a measurement problem, not a power problem. A null result cannot be read as no effect, and a significant coefficient cannot be read as a data-center effect. A better measure of metered data-center consumption would still be incomplete, because the capacity channel runs through expected load (Section 6.2). Answering the question requires both facility-level consumption and the load that PJM's forecast attributes to data centers.

6.4 Why the states differ

State differences are not ordered by regulatory structure. The capacity price coefficient is significant in Pennsylvania and Virginia and insignificant in Ohio and New Jersey (Table 5b). Ohio and Virginia faced the same capacity price in every delivery year except 2025/26 (Section 3.5), yet the coefficient is close to zero in restructured Ohio and significant in regulated Virginia,

whose residential price rose 3.6 percent between 2019 and 2025 against 11.7 percent in Ohio. Because the capacity signal was the same, the difference lies in how each state's prices respond to it. Restructured states, where supplier prices reset on a short cycle (Section 2.3), would be expected to respond most strongly, and Ohio does not. With one regulated state, this identifies Virginia against the other three, and the data do not indicate whether regulation, generation mix, or something else explains the difference.

New Jersey points away from data-center load. Its commercial sales fell between 2019 and 2025, so the proxy shows no sign of large new load, yet its real residential price rose 14.2 percent, the largest increase of the four states. Limited transmission capacity is the usual explanation for New Jersey's high prices (Section 2.4), and it is consistent with this pattern, but this analysis does not test it.

Other cost drivers, including the transmission congestion discussed in Section 2.4, plausibly raise prices independent of data centers. They lie outside this paper's tests, and the results should not be read as ranking them against the capacity price (Section 8).

6.5 The information-sector GDP sign

Information-sector GDP is significant in every pooled specification but negative: higher information-sector output is associated with lower residential prices (Tables 3 and 3b). A measure of data-center demand should not have this sign. The sign is also not consistent across states. It is positive but insignificant in Ohio, Virginia, and New Jersey, and negative in Pennsylvania (Table 5).

The likelier explanation is timing. Spending on data centers reaches state income accounts slowly, as facilities are built, equipped, and brought into service, while the capacity

price responds to expected load as soon as it is announced. Expectations of future growth can therefore raise prices before the information sector shows any matching output. This explanation is not tested here. The negative sign is best read as evidence about the proxy, not about data centers. It reinforces the conclusion of Section 6.3 that information-sector GDP does not isolate data-center activity.

7. Policy Implications

7.1 Require disclosure of large-facility electricity use

The central finding of this paper is a measurement gap, and the most direct response is to close it. No public source reports data-center electricity use by state and month (Section 3.6). The Lawrence Berkeley National Laboratory's national estimate rests on a bottom-up model built from limited public and proprietary data, and its authors note that very few companies report actual consumption (Shehabi et al., 2024). The recommendation is that facilities above a size threshold set by regulators report monthly electricity use to state regulators, with results published in aggregate by state or utility zone. Utilities meter every customer for billing, so the underlying data exist; what is missing is publication. Aggregation protects individual facilities, and the Berkeley Lab report points to anonymized, aggregated release as a workable model.

Disclosure sets no prices, assigns no costs, and blocks no construction. It makes the question answerable. With facility-level consumption by state and month, the proxy problem in Section 6.3 disappears, and the relationship between realized data-center load and residential prices can be tested directly rather than through commercial sales and information-sector GDP.

7.2 Approaches already in use

Some regulators address the cost side directly. In July 2025, the Public Utilities Commission of Ohio ordered AEP Ohio to file a data-center-specific tariff and to lift its moratorium on connecting new data centers, adopting a settlement that the Commission said protects other customers from the cost-shifting risk of underused investments made to serve data centers (Public Utilities Commission of Ohio, 2025). Tariffs of this kind aim to keep the cost of serving data centers with the facilities that cause it. They do not generate public data on consumption, and this paper does not evaluate their effect on residential prices. Disclosure and tariffs are complements: disclosure would let regulators observe whether tariffs work.

7.3 Forecasting and the capacity market

Because the capacity channel runs through expected load, transparency about the forecast matters as much as transparency about consumption. PJM began screening large-load requests more strictly before counting them in its 2026 forecast (PJM Interconnection, 2026a), and it is working through a stakeholder process on large load additions (PJM Interconnection, 2026e). Two additions would let outside researchers evaluate the channel: publishing how much forecast load comes from large loads and how much of it has been screened, and comparing forecast load with realized load after each delivery year. A persistent gap between the two is the condition under which Watten et al. (2026) find that the price-lowering effect of new demand reverses.

7.4 What to watch

Three developments will sharpen the evidence. First, the 2029/30 auction, scheduled for December 2026, would be the third consecutive auction to clear more than one percentage point below PJM's reserve target if it falls short again, the condition that requires a Reliability Backstop Auction under the current tariff (PJM Interconnection, 2026e). Second, the price limits

that capped the last three auctions run through the 2029/30 auction, so the first uncapped result comes later; PJM's own simulation of the 2028/29 auction without the cap produced a clearing price of $554.72 per megawatt-day against the $325.00 ceiling (PJM Interconnection, 2026e). Third, the sample ends seven months after the Dominion price separated from the RTO price, so each additional year of data will test whether the state differences in Table 5b persist.

8. Limitations

8.1 Measurement

No series in the panel measures data-center electricity use, so the central question cannot be answered directly (Section 3.6). Commercial sales and information-sector GDP are the closest proxies, and neither isolates data centers (Section 6.3). Other measurement choices also limit the results: quarterly GDP series are held constant within each quarter (Section 3.2), Pennsylvania's assignment to the MAAC zone is an approximation, Ohio and Virginia differ in capacity price only in 2025/26, which is represented by seven months (Section 3.5), and all prices are deflated with a national rather than a state-specific index (Section 3.3).

8.2 Inference

The analysis is descriptive, and its coefficients are associations, not effects. Natural gas is the only instrumented variable, and because its predicted value is estimated in a first stage, the second-stage standard errors do not account for that estimation error. The pooled model clusters standard errors by state, and with four clusters, cluster-robust inference can be unreliable, so the significance of the capacity price in Table 3 should be read cautiously (Section 4.4). The per-state estimates rely on 223 monthly observations with autocorrelation-robust standard errors, and the state-level capacity result is borderline in Pennsylvania ($p = .044$) and, in Virginia, once the

months of Dominion's price separation are excluded (p = .054) (Table 5b). A walk-forward forecast test finds out-of-sample predictive skill only in Ohio (Appendix D).

8.3 Scope

The sample is four states in one wholesale market, so the results do not generalize to other regions or market designs, and Virginia is the only regulated state (Section 2.3). The sample ends in December 2025, before any ceiling-level auction took effect, so later pass-through is outside it. Other cost drivers, including transmission congestion, are not tested and are not ranked against the capacity price (Section 6.4). The analysis also cannot say how much of the rise in capacity prices data centers explain, because it does not observe how much of PJM's forecast load comes from them. It shows that the capacity price tracks residential prices, and that the data needed to test the data-center link do not currently exist.

9. Conclusion

Residential electricity prices rose in real terms in all four states between 2005 and 2025, by 21.7 to 29.0 percent comparing the first and last twelve months of the sample (Section 5.1). The capacity price is the variable that tracks them. Its coefficient is positive, as expected, and it survives the functional forms tested, although it is significant in only two of the four states (Tables 3, 3b, and 5b). Natural gas prices, wholesale prices, and manufacturing GDP show no relationship once state and seasonal differences are controlled for.

The capacity price responds to expected load rather than consumption, which makes it a plausible channel for data centers: announced loads raise PJM's forecast before they draw any power. Evidence on realized load, by contrast, finds that new demand might spread fixed costs and lower average rates (Watten et al., 2026). The two findings describe different stages, and

which dominates depends on whether forecast load materializes. This analysis cannot settle that. Neither commercial sales nor information-sector GDP isolates data centers, and no public series measures their electricity use.

The available evidence is insufficient to determine whether data centers are contributing to higher residential electricity prices. Requiring large facilities to report their electricity use to state regulators, with the data published in aggregate, would allow for a more direct assessment of the relationship.

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Appendix

A. Variable Codebook

This appendix lists every variable that entered the final models reported in the Results section (Table A1). Each nominal dollar variable is deflated to real 2017 dollars using the BEA Personal Consumption Expenditures Price Index, applied identically across all four states. A smaller number of variables were assembled during the project but were not used in the final specifications. Those are noted at the end of this appendix along with the reason each was set aside, so the variable list here matches exactly what produced the numbers reported in the paper.

Table A1. Variables used in the final panel specifications.

Variable	Role	Unit	Source
ElecPrice	Dependent variable, all models	cents per kWh (2017 dollars)	EIA Electric Power Monthly
PredNG	Predicted natural gas price, first stage fitted value	dollars per Mcf (2017 dollars)	Constructed, control-function approach
RegHubLMP	Wholesale price at the state's regional PJM hub	dollars per MWh	PJM Data Miner 2
CapPrice	PJM capacity price, Base Residual Auction clearing price	dollars per MW-day (2017 dollars)	PJM Data Miner 2, unavailable before June 2007
ComSales	Commercial sector electricity sales, data center demand proxy	MWh	EIA Electric Power Monthly
ManufGDP	Manufacturing sector real GDP	millions, chained 2017 dollars	BEA Regional Economic Accounts
InfoGDP	Information sector real GDP, includes data processing and hosting	millions, chained 2017 dollars	BEA Regional Economic Accounts
HDD	Heating degree days, base 65 F	degree days	NOAA National Centers for Environmental Information
CDD	Cooling degree days, base 65 F	degree days	NOAA National Centers for Environmental Information
Time_Trend	Linear time trend, 1 for January 2005 through 252 for December 2025	integer	Constructed
State FE / Month FE	State and calendar month fixed effects	categorical	Constructed

Three variables were tested but excluded from the core specifications reported in the Results section. Industrial sales is discussed further, with its variance inflation factor, in Appendix B.

Industrial sector electricity sales was dropped because its value ranges barely overlap across the four states, so it behaves as a near-perfect state identifier rather than an independent economic signal. Total generation and the PJM RTO-wide hub price were dropped for the same reason, since each shares nearly all its variation with a variable already in the model, commercial sales and the regional hub price respectively.

A national data center construction spending series (U.S. Census Bureau, 2026) and a state data processing employment index were also assembled during the project as more direct data center proxies. Both were set aside because the construction series is only available at the national level and cannot support a state-level claim, and the employment series was suppressed for most states by the source agency for confidentiality reasons and was only usable for New Jersey. Commercial sales and information sector GDP remained the two data center proxies carried into the final models.

B. Full Regression Specifications

The pooled panel regression reported as Table 3 in Section 5.1 uses the nine explanatory variables listed in Appendix A, plus state and month fixed effects, estimated on all 892 monthly observations from June 2007 through December 2025 with standard errors clustered by state. Table B1 reports a broader version of the same regression that adds every candidate variable considered during the project, including the ones ultimately excluded, so the exclusion decisions in Appendix A can be checked directly against their coefficients, significance, and variance inflation factors.

Table B1. Full candidate variable regression, pooled across all four states, June 2007 through December 2025 (N = 892). State and month fixed effects included but not shown. Standard

errors clustered by state. Commercial sales are per MWh and GDP per million dollars, so those coefficients are in different units than in Table 3.

Variable	Coefficient	p-value	VIF	In core model
Predicted NG price	−0.0064	0.836	5.6	Yes
Regional hub LMP	+0.0009	0.823	1.9	Yes
Capacity price	+0.0032	0.003	1.3	Yes
Commercial sales	+0.0000003	0.183	3.0	Yes
Manufacturing GDP	+0.0000136	0.527	33.2	Yes
Information sector GDP	−0.0000346	< .001	11.2	Yes
Heating degree days	−0.0006	0.081	21.9	Yes
Cooling degree days	−0.0018	0.029	12.0	Yes
Time trend	+0.0021	0.623	4.2	Yes
Industrial sales	−0.0000002	0.546	79.8	No, state identifier artifact

Industrial sales carries a variance inflation factor above 79, far above the conventional concern threshold of 10, and its inclusion does not change the sign or significance of the capacity price coefficient. It is reported here for transparency and dropped from the core model used throughout the Results section. The capacity price coefficient is stable, small in magnitude, and significant whether industrial sales is included, as Table B1 shows (compare Table 3).

C. Machine Learning Robustness Check

A random forest model was fit on the same panel as a non-causal check on which variables carry the most predictive weight, independent of the linear regression's functional form assumptions. Permutation importance ranked industrial sales as overwhelmingly the strongest predictor of residential price, well ahead of every other variable in the model.

That result does not indicate a real economic effect. Industrial sales' predictive power comes from the same problem identified in Appendix B: its value ranges are almost entirely distinct across the four states, so a model can use it to identify which state an observation belongs to rather than to learn anything about the relationship between industrial activity and price. Dropping industrial sales from the random forest shifts the same artificial importance onto whichever remaining variable is most correlated with state identity, manufacturing GDP in one specification and the Virginia state indicator in another, confirming that the pattern is a labeling artifact rather than a signal about industrial demand. Neither the random forest nor the drop-one-out test that followed it changes the conclusions reported in Section 5.

D. Forecast Validation

A walk-forward backtest was used to test whether the model has genuine out-of-sample forecasting skill, as distinct from the in-sample explanatory regressions reported elsewhere. The model is fit on a rolling 60-month window, used to forecast 12 months ahead, and rolled forward every 6 months, producing 26 forecast origins per state. One specification, using the predicted natural gas price, capacity price, and commercial sales as controls, was applied identically across all four states so that the reported skill reflects one model rather than four separately tuned ones (Table D1).

Table D1. Walk-forward out-of-sample forecast skill, uniform specification applied identically across all four states.

State	Out-of-sample R-squared	MAE (cents/kWh)	Forecast origins	Interpretation
Ohio	0.184	0.465	26	Genuine skill
Pennsylvania	−0.546	0.741	26	No skill versus naive baseline
Virginia	−0.107	0.430	26	No skill versus naive baseline
New Jersey	−0.029	0.645	26	No skill versus naive baseline

Ohio is the only state where the model beats a naive baseline out of sample. An earlier version of this exercise tuned the natural gas measure and control set separately for each state rather than holding one specification fixed, and produced somewhat higher individual scores in Ohio and New Jersey. That version was set aside in favor of the single uniform specification reported here, since a model that changes shape from state to state is harder to justify and to apply going forward than one equation applied the same way everywhere. The conclusion is unchanged either way: genuine forecasting skill is present in Ohio and not reliably present in the other three states.

E. Correlation Matrices by State

Tables E1 through E4 report simple pairwise correlations among the five variables central to the paper's argument (residential price, capacity price, predicted natural gas price, commercial sales, and information sector GDP), computed separately for each state over June 2007 through December 2025. They illustrate the cross-state heterogeneity discussed throughout the Results section: the same variable can carry a very different relationship to price from one state to the next.

Table E1. Ohio, pairwise correlations, June 2007 through December 2025 (N = 223).

	ResPrice	**CapPrice**	**PredNG**	**ComSales**	**InfoGDP**
ResPrice	1.00	0.10	0.36	0.42	0.31
CapPrice	0.10	1.00	0.02	0.10	−0.13
PredNG	0.36	0.02	1.00	0.65	0.08
ComSales	0.42	0.10	0.65	1.00	0.32
InfoGDP	0.31	−0.13	0.08	0.32	1.00

Table E2. Pennsylvania, pairwise correlations, June 2007 through December 2025 (N = 223).

	ResPrice	**CapPrice**	**PredNG**	**ComSales**	**InfoGDP**
ResPrice	1.00	0.05	0.06	−0.11	0.29
CapPrice	0.05	1.00	0.04	0.35	−0.07
PredNG	0.06	0.04	1.00	0.44	−0.31
ComSales	−0.11	0.35	0.44	1.00	−0.49
InfoGDP	0.29	−0.07	−0.31	−0.49	1.00

Table E3. Virginia, pairwise correlations, June 2007 through December 2025 (N = 223).

	ResPrice	**CapPrice**	**PredNG**	**ComSales**	**InfoGDP**
ResPrice	1.00	0.23	0.33	0.31	0.20
CapPrice	0.23	1.00	0.05	0.03	−0.01
PredNG	0.33	0.05	1.00	0.19	−0.02
ComSales	0.31	0.03	0.19	1.00	0.92
InfoGDP	0.20	−0.01	−0.02	0.92	1.00

Table E4. New Jersey, pairwise correlations, June 2007 through December 2025 (N = 223).

	ResPrice	**CapPrice**	**PredNG**	**ComSales**	**InfoGDP**
ResPrice	1.00	0.43	0.66	0.46	−0.40
CapPrice	0.43	1.00	0.31	0.20	−0.56
PredNG	0.66	0.31	1.00	0.58	−0.33
ComSales	0.46	0.20	0.58	1.00	−0.27
InfoGDP	−0.40	−0.56	−0.33	−0.27	1.00

The capacity price is positively correlated with residential price in all four states, but weakly in Ohio (0.10) and Pennsylvania (0.05) and most strongly in New Jersey (0.43). Commercial sales and information-sector GDP are nearly collinear in Virginia (0.92) but not in the other states, which makes their separate relationships with price hard to distinguish there. Information-sector GDP is negatively correlated with residential price in New Jersey (−0.40) and positively in the

other three states (0.20 to 0.31). These differences are why the pooled regression in Appendix B relies on within-state variation through the fixed effects rather than on the raw pooled correlation.

F. Specifications Tested and Not Adopted

Two specifications were tested during the project and set aside. Both are reported here in full because a negative result from a reasonable specification is informative.

A continuous piecewise threshold model was fit to test whether the capacity price effect on residential price changes shape above some dollar threshold, rather than moving linearly throughout its range. The search identified a best-fitting threshold near $85 per megawatt-day, but neither the slope below that threshold nor the slope above it was statistically significant, at $p = 0.40$ and $p = 0.18$ respectively. The residual sum of squares improved by less than half of one percent relative to the linear specification, and the search curve across candidate thresholds was nearly flat, with two comparable local minima rather than one clear best fit. Together these point to weak identification rather than a genuine threshold, and the linear specification was kept.

Natural gas drilling rig counts were considered as an additional instrument for the predicted natural gas price, alongside the instrument set already used in the first stage. Adding rig counts was not adopted, to keep the instrument set consistent with the one already validated for joint relevance, rather than introduce a new instrument without repeating that validation across every state and period in the panel.

Earlier in the project, a data center construction spending series and a single national natural gas price instrumented through a separate two-stage least squares approach were also tested. Each is discussed in the relevant section of the paper: the data center spending series in Appendix A and

the first-stage natural gas specification that was adopted instead in Section 3.4.1, rather than repeated here.

G. Software

Excel, R, and Python were used in this analysis. Data from the sources in the References were compiled and deflated in Excel, which also produced the summary statistics in Table 1. The models were estimated in R and reproduced in Python (statsmodels for the panel regressions, linearmodels for the instrumental variables first stage, and scikit-learn for the random forest check), and the results were cross-checked against the Excel data.